

# PrimeCore Network Design

## *PrimeCore Minerals | Secure Multi Office Network*

PC-13 | Version 0.1 | 23 September 2026

Status: HQ addressing evidenced and wider design proposed

Owner: Network Design Owner | Approver: Project lead | Approval date: Pending

Document the HQ logical network, trust boundaries and monitoring path for PrimeCore's secure multi office project. This replaces the checklist's Health Centre label with the project's actual organisation.

## Logical architecture

The evidenced HQ firewall has upstream WAN 192.168.1.159/24 via DHCP, LAN 10.10.10.1/24 and VLAN interfaces on vtnet2. The WAN is a private lab upstream address. It must not be described as a public Internet address. Switch port mappings and physical links still require verification. [S1]

| VLAN and role     | Subnet and gateway  | Firewall interface |
|-------------------|---------------------|--------------------|
| 10 Employee wired | 192.168.10.0/24; .1 | vtnet2.10          |
| 20 Visitor        | 192.168.20.0/24; .1 | vtnet2.20          |
| 11 Employee Wi-Fi | 192.168.11.0/24; .1 | vtnet2.11          |
| 30 Printer        | 192.168.30.0/24; .1 | vtnet2.30          |
| 40 Video          | 192.168.40.0/24; .1 | vtnet2.40          |
| 41 CCTV           | 192.168.41.0/24; .1 | vtnet2.41          |
| 50 IoT            | 192.168.50.0/24; .1 | vtnet2.50          |
| 90 Server         | 192.168.90.0/24; .1 | vtnet2.90          |
| 91 Backup         | 192.168.91.0/24; .1 | vtnet2.91          |
| 99 Management     | 192.168.99.0/24; .1 | vtnet2.99          |

Confirmed endpoint placement: HQ-EMP-LAPTOP 192.168.11.100/24, gateway 192.168.11.1; HQ-MONITOR 192.168.90.60/24, gateway 192.168.90.1. The monitoring server is on the server VLAN, even though administrators use it for security work.

## Boundary rationale

Separate staff, visitors, devices, backup and administration so each boundary can enforce required flows. VLANs provide separation only when trunk configuration and routing policy are correct. Same-subnet traffic may bypass the firewall. Verify endpoint or switching controls for required isolation within a VLAN.

## Traffic design and operational dependencies

| Origin                 | Allowed destination and purpose                                  | Required restriction                                         |
|------------------------|------------------------------------------------------------------|--------------------------------------------------------------|
| Employee               | Approved DNS, time, identity, files, print and Internet services | Deny management and backup except specific approved service  |
| Visitor                | Approved DNS and Internet access                                 | Deny all project internal and lab upstream management ranges |
| IoT and CCTV           | Named controllers and approved updates                           | Deny lateral user and management access                      |
| Backup service         | Explicit protected hosts and repository ports                    | Ordinary clients cannot delete recovery copies               |
| Administrator endpoint | Named devices and management ports                               | Dedicated identities and limited source addresses            |
| HQ-FW                  | HQ-MONITOR UDP 514                                               | Limit collector acceptance to actual firewall source         |

These are proposed flow requirements, not a complete current allow list. Register actual directory, file, print, DNS and NTP servers before implementation. Validate both IPv4 and IPv6 or explicitly document a controlled IPv6-disabled scope. Confirm DHCP scopes, exclusions, reservations and lease attribution; do not invent them from gateway addresses.

## Monitoring data path

HQ-FW sends syslog to HQ-MONITOR over UDP 514. Rsyslog writes /var/log/pfsense.log. Alloy reads the file and forwards to local Loki. Grafana queries job=pfsense and site=brisbane-hq and routes alerts through SMTP. UDP forwarding needs restricted access, local retention and loss or health checks. Listener presence alone does not prove network permissions. [S2]

## Multi office and cloud extension

Record each additional site's subnets, gateway, WAN arrangement, link capacity and permitted cross-site flows before assigning routes. Use non-overlapping ranges and an approved encrypted site-to-site design; branch addressing and operational tunnels are not evidenced here. PC-05 proposes Azure 10.60.0.0/16 and P2S 172.30.250.0/24 independently of HQ. Reaching HQ from Azure would require a separate validated route and site-to-site design.

## Validation and design handover

Capture an annotated EVE topology with actual port-to-port links, VLAN trunks and access ports. Test one allowed and one denied flow per boundary, DNS and time, gateway failure impact, throughput and log attribution. Existing T01 proves only the stated employee ICMP block. Record observed bandwidth and peak traffic before selecting production hardware. Duane's assigned network design work should be reconciled with this HQ baseline; completion or authorship is not inferred from assignment.

## References and linked artefacts

[S1] HQ\_Network\_Addressing\_Artefact.md, project deliverables, evidence dated 13 September 2026.

[S2] Security\_Alerting\_Implementation\_Artefact.md, project deliverables, 13 September 2026. Credits Jared Williams for the original monitoring foundation and Adam Donovan for subsequent remediation, alerting and validation.

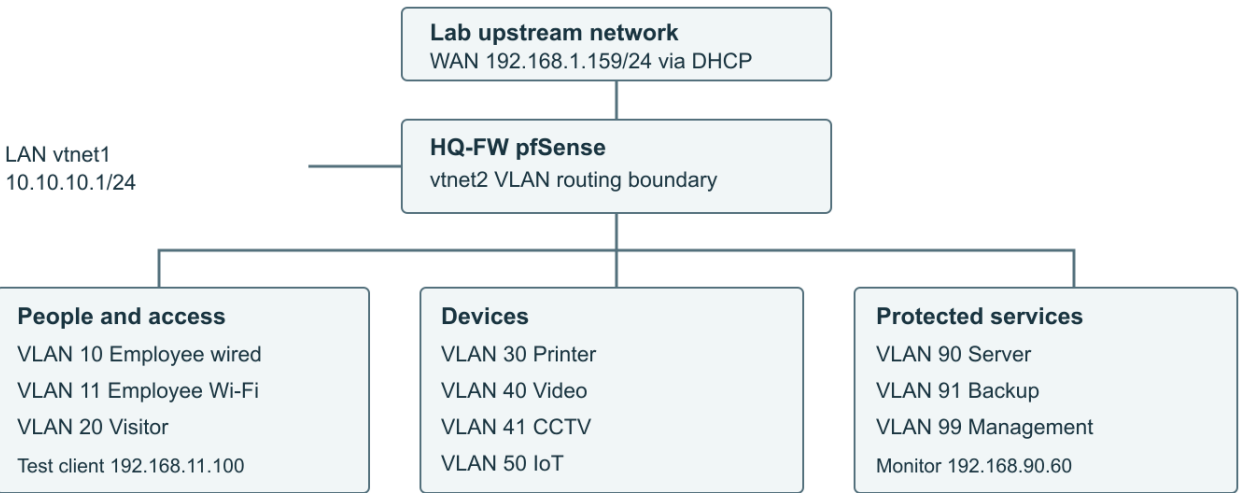
[S3] Security\_Test\_Results.md, project deliverables, 13 September 2026, tests T01 to T05 and their evidence limitations.

# Logical topology and trust boundaries

The figure records logical network relationships. It is not a physical cabling or verified switch-port map. The Azure prototype remains a separate proposed design.

## PrimeCore HQ logical network

Addressing observed 13 September 2026; lines represent logical interfaces



Each VLAN uses 192.168.<VLAN>.0/24 with gateway .1  
R06 test: 192.168.11.100 to 192.168.99.1 ICMP blocked and logged

**Verified monitoring path**  
HQ-FW syslog → rsyslog → /var/log/pfsense.log → Alloy → Loki → Grafana  
UDP 514 to HQ-MONITOR; received R06 email alerts through SMTP TCP 587

**Proposed Azure prototype — separate design**  
Remote client pool 172.30.250.0/24 → Azure VPN gateway → VNet 10.60.0.0/16  
Private workload 10.60.10.10; no HQ tunnel or branch connectivity claimed

Switch ports, physical links, branch topology and complete policy enforcement require verification.

Sources: S1 for HQ addressing; S2 and S3 for the monitoring and R06 test path. Proposed Azure values correspond to PC-05 and PC-15. The editable SVG is supplied alongside this document.

## Screenshot evidence appendix

These figures record selected laboratory observations. They support the tested functions described in this artefact. Proposed policies, untested controls and organisational approval remain separate from the pictured results.

Figure E1 HQ firewall interfaces

```
WAN (wan)          -> vtnet0      -> v4/DHCP4: 192.168.1.159/24
LAN (lan)          -> vtnet1      -> v4: 10.10.10.1/24
EMPLOYEE_VLAN (opt1) -> vtnet2.10 -> v4: 192.168.10.1/24
VISITOR_VLAN (opt2) -> vtnet2.20 -> v4: 192.168.20.1/24
EMPLOYEE_WIFI_VLAN (opt3) -> vtnet2.11 -> v4: 192.168.11.1/24
PRINTER_VLAN (opt4) -> vtnet2.30 -> v4: 192.168.30.1/24
VIDEO_VLAN (opt5)   -> vtnet2.40 -> v4: 192.168.40.1/24
CCTV_VLAN (opt6)    -> vtnet2.41 -> v4: 192.168.41.1/24
IOT_VLAN (opt7)     -> vtnet2.50 -> v4: 192.168.50.1/24
SERVER_VLAN (opt8)  -> vtnet2.90 -> v4: 192.168.90.1/24
BACKUP_VLAN (opt9)  -> vtnet2.91 -> v4: 192.168.91.1/24
MANAGEMENT_VLAN (opt10) -> vtnet2.99 -> v4: 192.168.99.1/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell

9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + Netgate pfSense Plus tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: █
```

The interface register shows the recorded VLAN gateways. Interface presence alone does not demonstrate that every switch port or access rule has been tested.

Source evidence: evidence/HQ\_FW\_Interface\_Map.png

## Screenshot evidence continued

Figure E2 Backup access port configured

```
labadmin@HQ-SW-OVS:~$ ip -br link
lo                UNKNOWN    00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
ens3              UP          00:50:00:00:06:00 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens4              UP          00:50:00:00:06:01 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens5              UP          00:50:00:00:06:02 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens6              UP          00:50:00:00:06:03 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens7              UP          00:50:00:00:06:04 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens8              UP          00:50:00:00:06:05 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens9              UP          00:50:00:00:06:06 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens10             UP          00:50:00:00:06:07 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens11             UP          00:50:00:00:06:08 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens12             UP          00:50:00:00:06:09 <BROADCAST,MULTICAST,UP,LOWER_UP>
ens13             UP          00:50:00:00:06:0a <BROADCAST,MULTICAST,UP,LOWER_UP>
ens14             UP          00:50:00:00:06:0b <BROADCAST,MULTICAST,UP,LOWER_UP>
ovs-system        DOWN       d6:c1:f3:1c:2c:fd <BROADCAST,MULTICAST>
mgmt99           UNKNOWN    0e:ea:f3:4b:be:82 <BROADCAST,MULTICAST,UP,LOWER_UP>
br0              DOWN       00:50:00:00:06:00 <BROADCAST,MULTICAST>
labadmin@HQ-SW-OVS:~$ sudo ovs-vsctl --may-exist add-port br0 ens14 -- set Port ens14 vlan_mode=access tag=91
[sudo: authenticate] Password:
labadmin@HQ-SW-OVS:~$ sudo ovs-vsctl list Port ens14
_uuid            : 3866d9a7-301b-46c0-a277-83d6292896f6
bond_active_slave : []
bond_downdelay   : 0
bond_fake_iface  : false
bond_mode        : []
bond_updelay     : 0
cvlans           : []
external_ids     : {}
fake_bridge      : false
interfaces       : [c2935839-16cf-4842-a7c4-bc11a3b09e50]
lacp             : []
mac              : []
name             : ens14
other_config     : {}
protected        : false
qos              : []
rstp_statistics  : {}
rstp_status      : {}
statistics       : {}
status           : {}
tag              : 91
trunks           : []
vlan_mode        : access
labadmin@HQ-SW-OVS:~$
```

ens14 has tag 91 and vlan\_mode access. Untagged server traffic is assigned to VLAN 91 by the switch.

Source evidence: evidence/Backup\_Server/B08\_b3ab8c42-1306-47e7-88dc-825aa10ec8cf.png

## Screenshot evidence continued

Figure E3 Successful DNS query

```
labadmin@HQ-BACKUP:~$ sudo netplan generate
labadmin@HQ-BACKUP:~$ sudo netplan try
Do you want to keep these settings?

Press ENTER before the timeout to accept the new configuration

Changes will revert in 119 seconds
Configuration accepted.
labadmin@HQ-BACKUP:~$ resolvectl status ens3
Link 2 (ens3)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    DNS Servers: 192.168.90.10
    Default Route: yes
labadmin@HQ-BACKUP:~$ resolvectl query ubuntu.com
ubuntu.com: 185.125.190.29 -- link: ens3
            185.125.190.20 -- link: ens3
            185.125.190.21 -- link: ens3
            2620:2d:4000:1::28 -- link: ens3
            2620:2d:4000:1::26 -- link: ens3
            2620:2d:4000:1::27 -- link: ens3

-- Information acquired via protocol DNS in 30.2ms.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: network
labadmin@HQ-BACKUP:~$ _
```

The configuration is accepted and ubuntu.com resolves through ens3 with DNS server 192.168.90.10.

Source evidence: evidence/Backup\_Server/B25\_952a7db0-21dd-4e1b-bbc2-c1475a172081.png